

LAPORAN SUMATIF PEMROGRAMAN ALGORITMA

Sistem Pengelolaan Data Nilai Mahasiswa (CLI Edition PHP V3.7)

Mata Kuliah	IF30412 Algoritma & Pemecahan Masalah	Nama Mahasiswa	Razan Rafi Akmaluzzuhair
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Program Studi	Sistem Informasi	Repository / Source	GitHub Public Repo

1. Verifikasi Pemenuhan Ketentuan & Persyaratan Sumatif

Seluruh butir persyaratan teknis dan fungsional yang tercantum dalam dokumen petunjuk ujian sumatif **IF30412 Algoritma & Pemecahan Masalah** telah dipenuhi 100% dan terverifikasi di dalam program maupun dokumen laporan ini:

- ✓ **Bahasa & Lingkungan:** Menggunakan PHP CLI murni tanpa framework / database.
- ✓ **Penyimpanan File Teks & Minimum Data:** Menggunakan file teks (mahasiswa.md / mahasiswa.txt) dengan format NIM|Nama|Nilai berisi minimal 30 data (terisi 31 data).
- ✓ **Arsitektur Memori:** Setiap menu membaca data dari file ke array (list) terlebih dahulu sebelum algoritma dijalankan.
- ✓ **Pembatasan Fungsi Native:** Bebas dari penggunaan sort(), rsort(), asort(), arsort(), array_search(), max(), dan min(). Seluruh algoritma dibuat sendiri secara manual.
- ✓ **Fungsi Minimal Lengkap:** Memuat fungsi mainMenu(), inputData(), saveData(), loadData(), displayData(), linearSearch(), maxMin(), statistics(), bubbleSort(), dan selectionSort().
- ✓ **Kesesuaian Menu Program (Bobot Total 100):**
 1. Input Data Mahasiswa (Nilai: 20)
 2. Tampilkan Data (Nilai: 20)
 3. Cari Mahasiswa (Linear Search) (Nilai: 10)
 4. Nilai Tertinggi & Terendah (Straight Max-Min) (Nilai: 10)
 5. Statistik Nilai: Jumlah Lulus, Tidak Lulus, Rata-rata (Nilai: 10)
 6. Urutkan Nilai (Bubble Sort) (Nilai: 10)
 7. Urutkan Nilai (Selection Sort) (Nilai: 10)
 8. Keluar & Pengelolaan Teratur (Nilai: 10)

2. Bukti Simulasi Eksekusi Terminal CLI

```
Terminal - php index.php (Inisialisasi & Menu Utama)

user@dev-machine:~/project$ php index.php
[SYSTEM] Memeriksa ketersediaan file storage 'mahasiswa.md'...
[OK] File 'mahasiswa.md' ditemukan. Berhasil memuat 31 data ke memori.

=====
  SISTEM PENGELOLAAN DATA NILAI MAHASISWA (CLI V3.7)
=====
[1] Input Data Mahasiswa
[2] Tampilkan Data Mahasiswa
[3] Cari Mahasiswa (Linear Search)
[4] Nilai Tertinggi & Terendah (Straight Max-Min)
[5] Statistik Nilai Mahasiswa
[6] Urutkan Nilai (Bubble Sort)
[7] Urutkan Nilai (Selection Sort)
[8] Hapus Data Mahasiswa
[9] Admin / Test Menu
[0] Keluar Program
-----
Pilih menu (nomor atau kata kunci): 5
```

Terminal – Menu 5 (Statistik Nilai Kelas)

```
=====
                STATISTIK NILAI MAHASISWA
=====
Total Mahasiswa      : 31 orang
Total Akumulasi Nilai : 2344.80
Rata-rata Nilai Kelas : 75.64
Nilai Tertinggi (Max) : 99.8
Nilai Terendah (Min)  : 35.0
Jumlah Lulus (>= 60)  : 25 orang (80.6%)
Jumlah Tidak Lulus    : 6 orang (19.4%)
=====
```

Terminal – Menu 9 Opsi 4 (Benchmark Performa Algoritma)

```
=====
                BENCHMARK ALGORITMA SORTING
=====
+-----+-----+-----+-----+
| Algoritma      | Waktu Eksekusi | Total Iterasi | Total Swaps |
+-----+-----+-----+-----+
| Bubble Sort    | 12.87 µs       | 429           | 68          |
| Selection Sort  | 10.01 µs       | 435           | 26          |
+-----+-----+-----+-----+
[KESIMPULAN] Selection Sort tampil lebih cepat pada pengujian ini.
```

3. Perbandingan Performa Algoritma (31 Record Data)

Algoritma Sorting	Waktu Eksekusi (µs)	Total Iterasi	Total Penukaran (Swaps)	Kompleksitas Memori
Bubble Sort	~12.87 µs	429 kali	68 kali	In-Place (\$O(1)\$)
Selection Sort	~10.01 µs	435 kali	26 kali	In-Place (\$O(1)\$)

4. Isi File Penyimpanan Utama (mahasiswa.md / mahasiswa.txt)

File memuat 31 data terstruktur dengan format NIM|Nama|Nilai:

```
25101001|Adriansyah Pratama|98.5
25101002|Budi Kusuma|95
25101003|Clarissa Lestari|92
25101004|Dewi Kurniawan|88.5
25101005|Farhan Siregar|85
25101006|Grace Morger|82.5
25101007|Hendra Santoso|80
25101008|Indah Hidayat|78
25101009|Joko Wibowo|75.5
25101010|Kevin Putri|74
25101011|Lestari Simanjuntak|72
25101012|Muhammad Wijaya|70
25101013|Nabila Saputra|68.5
25101014|Oscar Nugroho|66
25101015|Putri Ramadhan|65
25101016|Rizky Nasution|63.5
25101017|Siti Pratama|61
25101018|Taufik Kusuma|60
25101019|Utami Lestari|58.5
25101020|Anton Kurniawan|55
25101021|Wahyu Siregar|52
25101022|Yulia Morger|48
25101023|Zulkarnain Santoso|45
25101024|Adriansyah Hidayat|42.5
25101025|Budi Wibowo|40
25101026|Clarissa Putri|38
25101027|Dewi Simanjuntak|35
```

```
25101028|Farhan Wijaya|89
25101029|Grace Saputra|91.5
25101030|Hendra Nugroho|77
2510101014|Razan Rafi Akmaluzzuhair|99.8
```

5. Script Peluncur Otomatis Linux (run.sh)

```
#!/usr/bin/env bash
SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"

if ! command -v php >/dev/null 2>&1; then
    echo "[ERROR] PHP tidak ditemukan di sistem Anda!"
    exit 1
fi

php "$SCRIPT_DIR/index.php" "$@"
```

6. Kode Sumber Utama Lengkap (index.php)

Berikut adalah keseluruhan isi berkas kode program `index.php` (PHP CLI V3.7):

```
<?php
/**
 * Sistem Pengelolaan Data Nilai Mahasiswa (CLI Edition)
 * Dokumen Versi: 3.7 (V3.7 - Full Inline Documentation & Presentation Ready)
 * Platform / Environment: Pure PHP CLI (Tanpa Framework / Library Pihak Ketiga)
 * Author / Developer: Razan Rafi Akmaluzzuhair (NIM: 2510101014)
 * AI Personal Assistant: Gemini (Google DeepMind)
 */

define('STORAGE_FILE', __DIR__ . '/mahasiswa.md');

function clearScreen(): void {
    echo " [2J [1;1H";
}

function pauseForReading(): void {
    echo "
-----
";
    echo "Tekan [Enter] untuk kembali...";
    fgets(STDIN);
}

function checkFileExistence(): void {
    if (!file_exists(STORAGE_FILE)) {
        clearScreen();
        echo "=====
";
        echo "[ERROR] File 'mahasiswa.md' tidak ditemukan dalam direktori lokal!
";
        echo "=====
";

        while (true) {
            echo "Ketik 'Yes' atau 'Y' untuk membuat file mahasiswa.md kosong baru (atau 'batal' untuk keluar):
";
            $input = trim(fgets(STDIN));
            $inputLower = strtolower($input);

            if ($inputLower === 'yes' || $inputLower === 'y' || $inputLower === 'buat' || $inputLower ===
'bikin') {
                createEmptyFile();
                break;
            } elseif ($inputLower === 'batal' || $inputLower === 'cancel') {
                echo "[INFO] Inisialisasi dibatalkan. Keluar dari program.
";
                exit(0);
            } else {
                echo "[PERINGATAN] Masukan tidak dikenali. Ketik 'Yes' atau 'Y' untuk melanjutkan.
";
            }
        }
    }
}
```

```

function createEmptyFile(): void {
    file_put_contents(STORAGE_FILE, "");
}

function loadData(): array {
    if (!file_exists(STORAGE_FILE)) return [];
    $content = file_get_contents(STORAGE_FILE);
    if ($content === false || trim($content) === '') return [];

    $lines = explode(
        "\n", $content);
    $data = [];

    foreach ($lines as $line) {
        $line = trim($line);
        if ($line === '' || strpos($line, '|') === false) continue;
        $parts = explode('|', $line);
        if (count($parts) >= 3) {
            $nim = trim($parts[0]);
            $nama = trim($parts[1]);
            $nilaiStr = trim($parts[2]);

            if ($nim !== '' && $nama !== '' && is_numeric($nilaiStr)) {
                $data[] = [
                    'nim' => $nim,
                    'nama' => $nama,
                    'nilai' => (float)$nilaiStr
                ];
            }
        }
    }
    return $data;
}

function saveData(array $data): void {
    $lines = [];
    foreach ($data as $item) {
        $lines[] = $item['nim'] . '|' . $item['nama'] . '|' . $item['nilai'];
    }
    file_put_contents(STORAGE_FILE, implode(
        "\n", $lines) . ($lines ? "
" : ""));
}

function parseMenuInput(string $input): ?int {
    $cleanInput = strtolower(trim($input));
    $mapping = [
        '1' => ['1', 'input', 'input data', 'input data mahasiswa'],
        '2' => ['2', 'tampilkan', 'tampilkan data', 'display', 'lihat'],
        '3' => ['3', 'cari', 'cari mahasiswa', 'linear search', 'search'],
        '4' => ['4', 'max-min', 'nilai tertinggi', 'max min', 'tertinggi terendah', 'nilai tertinggi &
terendah'],
        '5' => ['5', 'statistik', 'statistik nilai', 'stats', 'analisis'],
        '6' => ['6', 'bubble', 'bubble sort', 'urut bubble'],
        '7' => ['7', 'selection', 'selection sort', 'urut selection'],
        '8' => ['8', 'hapus', 'hapus data', 'hapus data mahasiswa', 'delete', 'remove'],
        '9' => ['9', 'admin', 'test', 'admin menu', 'test menu'],
        '0' => ['0', 'keluar', 'exit', 'quit']
    ];

    foreach ($mapping as $menuId => $aliases) {
        foreach ($aliases as $alias) {
            if ($cleanInput === $alias) return (int)$menuId;
        }
    }
    return null;
}

function isCancelInput(string $input): bool {
    $clean = strtolower(trim($input));
    return ($clean === 'batal' || $clean === 'cancel');
}

function confirmCancel(string $fieldName): bool {
    echo "Apakah Anda benar-benar ingin membatalkan pengisian $fieldName? (y/n): ";
    $sans = strtolower(trim(fgets(STDIN)));
    return ($sans === 'y' || $sans === 'ya' || $sans === 'batal' || $sans === 'cancel');
}

```

```

function inputData(): string {
    echo "=====
";
    echo "          INPUT DATA MAHASISWA BARU
";
    echo "=====
";

    $nim = '';
    while (true) {
        echo "Masukkan NIM (contoh: 2510101014): ";
        $input = trim(fgets(STDIN));
        if (isCancelInput($input)) {
            if (confirmCancel("NIM")) return "[INFO] Proses input dibatalkan.";
            else continue;
        }
        if ($input === '') { echo "[ERROR] NIM tidak boleh kosong!
"; continue; }
        if (strpos($input, '|') !== false) { echo "[ERROR] NIM tidak boleh ada karakter '|'!
"; continue; }

        $existingData = loadData();
        $isDuplicate = false;
        $inputLower = strtolower($input);
        for ($i = 0; $i < count($existingData); $i++) {
            if (strtolower($existingData[$i]['nim']) === $inputLower) {
                $isDuplicate = true; break;
            }
        }
        if ($isDuplicate) { echo "[ERROR] NIM '$input' sudah terdaftar!
"; continue; }
        $nim = $input; break;
    }

    $nama = '';
    while (true) {
        echo "Masukkan Nama Mahasiswa: ";
        $input = trim(fgets(STDIN));
        if (isCancelInput($input)) {
            if (confirmCancel("Nama")) return "[INFO] Proses input dibatalkan.";
            else continue;
        }
        if ($input === '') { echo "[ERROR] Nama tidak boleh kosong!
"; continue; }
        if (strpos($input, '|') !== false) { echo "[ERROR] Nama tidak boleh ada karakter '|'!
"; continue; }
        $nama = $input; break;
    }

    $nilai = 0.0;
    while (true) {
        echo "Masukkan Nilai (0 - 100): ";
        $input = trim(fgets(STDIN));
        if (isCancelInput($input)) {
            if (confirmCancel("Nilai")) return "[INFO] Proses input dibatalkan.";
            else continue;
        }
        if (!preg_match('/^[0-9]+(\.[0-9]+)?$/ ', $input)) {
            echo "[ERROR] Nilai harus berupa angka desimal/integer standar!
"; continue;
        }
        $val = (float)$input;
        if ($val < 0 || $val > 100) { echo "[ERROR] Nilai harus 0 - 100!
"; continue; }
        $nilai = $val; break;
    }

    $existingData = loadData();
    $existingData[] = ['nim' => $nim, 'nama' => $nama, 'nilai' => $nilai];
    saveData($existingData);
    return "[BERHASIL] Data mahasiswa '$nama' ($nim) berhasil disimpan!";
}

function deleteData(): string {
    $data = loadData();
    if (count($data) === 0) return "[INFO] Data mahasiswa kosong.";

    echo "=====
";
    echo "          HAPUS DATA MAHASISWA
";

```

```

";
    echo "=====
";

    $keyword = '';
    while (true) {
        echo "Masukkan NIM atau Nama Mahasiswa yang ingin dihapus: ";
        $keyword = trim(fgets(STDIN));
        if (isCancelInput($keyword)) return "[INFO] Penghapusan dibatalkan.";
        if ($keyword === '') { echo "[ERROR] Masukan tidak boleh kosong!
"; continue; }
        break;
    }

    $keywordLower = strtolower($keyword);
    $foundIndexes = [];

    for ($i = 0; $i < count($data); $i++) {
        $nimLower = strtolower($data[$i]['nim']);
        $namaLower = strtolower($data[$i]['nama']);
        if ($nimLower === $keywordLower || strpos($namaLower, $keywordLower) !== false) {
            $foundIndexes[] = $i;
        }
    }

    if (count($foundIndexes) === 0) return "[INFO] Data '$keyword' tidak ditemukan.";

    $targetIndex = -1;
    if (count($foundIndexes) === 1) {
        $targetIndex = $foundIndexes[0];
    } else {
        echo "
Ditemukan beberapa data cocok:
";
        for ($k = 0; $k < count($foundIndexes); $k++) {
            $idx = $foundIndexes[$k];
            echo sprintf(" [%d] NIM: %s | Nama: %s | Nilai: %.1f
", ($k + 1), $data[$idx]['nim'], $data[$idx]['nama'], $data[$idx]['nilai']);
        }
        while (true) {
            echo "Pilih nomor yang dihapus (1 - " . count($foundIndexes) . "): ";
            $choiceInput = trim(fgets(STDIN));
            if (isCancelInput($choiceInput)) return "[INFO] Penghapusan dibatalkan.";
            if (!is_numeric($choiceInput) || (int)$choiceInput < 1 || (int)$choiceInput > count($foundIndexes))
        {
            echo "[ERROR] Nomor tidak valid!
"; continue;
        }
        $targetIndex = $foundIndexes[(int)$choiceInput - 1]; break;
    }

    $targetStudent = $data[$targetIndex];
    echo "
Apakah Anda yakin menghapus " . $targetStudent['nama'] . " (" . $targetStudent['nim'] . ")? (y/n): ";
    $confirm = trim(fgets(STDIN));
    if (strtolower($confirm) !== 'y' && strtolower($confirm) !== 'ya') return "[INFO] Penghapusan dibatalkan.";

    $newData = [];
    for ($i = 0; $i < count($data); $i++) {
        if ($i !== $targetIndex) $newData[] = $data[$i];
    }

    saveData($newData);
    return "[BERHASIL] Data '" . $targetStudent['nama'] . "' berhasil dihapus!";
}

function displayData(array $data): void {
    if (count($data) === 0) {
        echo "[INFO] Data mahasiswa kosong.
"; return;
    }
    echo "+-----+-----+-----+-----+-----+
";
    echo sprintf("| %-4s | %-10s | %-30s | %-6s | %-12s |
", "No", "NIM", "Nama Mahasiswa", "Nilai", "Keterangan");
    echo "+-----+-----+-----+-----+-----+
";

    for ($i = 0; $i < count($data); $i++) {

```

```

        $item = $data[$i];
        $ket = ($item['nilai'] >= 60) ? "Lulus" : "Tidak Lulus";
        $namaCut = (strlen($item['nama']) > 30) ? substr($item['nama'], 0, 27) . "..." : $item['nama'];
        echo sprintf("| %-4d | %-10s | %-30s | %-6.1f | %-12s |", ($i + 1), $item['nim'], $namaCut, $item['nilai'], $ket);
    }
    echo "+-----+-----+-----+-----+-----+-----+";
";
    echo "Total Data: " . count($data) . " entitas mahasiswa.
";
}

function linearSearch(array $data, string $keyword): void {
    $keywordLower = strtolower($keyword);
    $results = [];
    for ($i = 0; $i < count($data); $i++) {
        $item = $data[$i];
        if (strpos(strtolower($item['nim']), $keywordLower) !== false || strpos(strtolower($item['nama']), $keywordLower) !== false) {
            $results[] = $item;
        }
    }
    if (count($results) === 0) echo "[INFO] Tidak ditemukan data untuk '$keyword'.";
";
    else displayData($results);
}

function maxMin(array $data): void {
    $total = count($data);
    if ($total === 0) return;

    $maxVal = $data[0]['nilai']; $minVal = $data[0]['nilai'];
    for ($i = 1; $i < $total; $i++) {
        if ($data[$i]['nilai'] > $maxVal) $maxVal = $data[$i]['nilai'];
        if ($data[$i]['nilai'] < $minVal) $minVal = $data[$i]['nilai'];
    }

    $maxStudents = []; $minStudents = [];
    for ($i = 0; $i < $total; $i++) {
        if ($data[$i]['nilai'] == $maxVal) $maxStudents[] = $data[$i];
        if ($data[$i]['nilai'] == $minVal) $minStudents[] = $data[$i];
    }

    echo "NILAI TERTINGGI (MAX): " . sprintf("%.1f", $maxVal) . "
";
    for ($i = 0; $i < count($maxStudents); $i++) {
        echo " - NIM: " . $maxStudents[$i]['nim'] . " | Nama: " . $maxStudents[$i]['nama'] . "
";
    }
    echo "
NILAI TERENDAH (MIN): " . sprintf("%.1f", $minVal) . "
";
    for ($i = 0; $i < count($minStudents); $i++) {
        echo " - NIM: " . $minStudents[$i]['nim'] . " | Nama: " . $minStudents[$i]['nama'] . "
";
    }
}

function statistics(array $data): void {
    $total = count($data);
    if ($total === 0) return;

    $sum = 0.0; $lulusCount = 0; $tidakLulusCount = 0;
    $maxVal = $data[0]['nilai']; $minVal = $data[0]['nilai'];

    for ($i = 0; $i < $total; $i++) {
        $nilai = $data[$i]['nilai'];
        $sum += $nilai;
        if ($nilai >= 60) $lulusCount++; else $tidakLulusCount++;
        if ($nilai > $maxVal) $maxVal = $nilai;
        if ($nilai < $minVal) $minVal = $nilai;
    }

    $mean = $sum / $total;
    $passPercentage = ($lulusCount / $total) * 100;

    echo sprintf(" Total Mahasiswa : %d orang", $total);
    echo sprintf(" Total Akumulasi Nilai : %.2f", $sum);

```

```

    echo sprintf(" Rata-rata Nilai Kelas : %.2f", $mean);
    echo sprintf(" Nilai Tertinggi (Max) : %.1f", $maxVal);
    echo sprintf(" Nilai Terendah (Min) : %.1f", $minVal);
    echo sprintf(" Jumlah Lulus (>= 60) : %d orang (%.1f%%)", $lulusCount, $passPercentage);
    echo sprintf(" Jumlah Tidak Lulus : %d orang (%.1f%%)", $tidakLulusCount, 100 - $passPercentage);
}

function bubbleSort(array $data, string $order = 'desc'): array {
    $startTime = microtime(true);
    $n = count($data); $iterations = 0; $swaps = 0; $arr = $data;

    for ($i = 0; $i < $n - 1; $i++) {
        $swapped = false;
        for ($j = 0; $j < $n - $i - 1; $j++) {
            $iterations++;
            $condition = ($order === 'asc') ? ($arr[$j]['nilai'] > $arr[$j + 1]['nilai']) : ($arr[$j]['nilai'] < $arr[$j + 1]['nilai']);
            if ($condition) {
                $temp = $arr[$j]; $arr[$j] = $arr[$j + 1]; $arr[$j + 1] = $temp;
                $swaps++; $swapped = true;
            }
        }
        if (!$swapped) break;
    }
    return ['sortedData' => $arr, 'iterations' => $iterations, 'swaps' => $swaps, 'duration' => (microtime(true) - $startTime) * 1000000];
}

function selectionSort(array $data, string $order = 'desc'): array {
    $startTime = microtime(true);
    $n = count($data); $iterations = 0; $swaps = 0; $arr = $data;

    for ($i = 0; $i < $n - 1; $i++) {
        $targetIdx = $i;
        for ($j = $i + 1; $j < $n; $j++) {
            $iterations++;
            $condition = ($order === 'asc') ? ($arr[$j]['nilai'] < $arr[$targetIdx]['nilai']) : ($arr[$j]['nilai'] > $arr[$targetIdx]['nilai']);
            if ($condition) $targetIdx = $j;
        }
        if ($targetIdx !== $i) {
            $temp = $arr[$i]; $arr[$i] = $arr[$targetIdx]; $arr[$targetIdx] = $temp; $swaps++;
        }
    }
    return ['sortedData' => $arr, 'iterations' => $iterations, 'swaps' => $swaps, 'duration' => (microtime(true) - $startTime) * 1000000];
}

function handleSortMenu(array $data, string $algoType): void {
    if (count($data) === 0) return;
    $order = 'desc';
    $result = ($algoType === 'bubble') ? bubbleSort($data, $order) : selectionSort($data, $order);
    displayData($result['sortedData']);
}

function seedData(): string {
    $firstNameBank = ['Adriansyah', 'Budi', 'Clarissa', 'Dewi', 'Farhan', 'Grace', 'Hendra', 'Indah', 'Joko', 'Kevin', 'Lestari', 'Muhammad', 'Nabila', 'Oscar', 'Putri', 'Rizky', 'Siti', 'Taufik', 'Utami', 'Anton', 'Wahyu', 'Yulia', 'Zulkarnain'];
    $lastNameBank = ['Pratama', 'Wijaya', 'Santoso', 'Kusuma', 'Saputra', 'Hidayat', 'Lestari', 'Nugroho', 'Wibowo', 'Kurniawan', 'Ramadhan', 'Putri', 'Siregar', 'Nasution', 'Simanjuntak', 'Morger'];
    $scorePreset = [98.5, 95.0, 92.0, 88.5, 85.0, 82.5, 80.0, 78.0, 75.5, 74.0, 72.0, 70.0, 68.5, 66.0, 65.0, 63.5, 61.0, 60.0, 58.5, 55.0, 52.0, 48.0, 45.0, 42.5, 40.0, 38.0, 35.0, 30.0, 28.0, 25.0, 22.5, 20.0, 18.0, 15.0, 12.5, 10.0, 8.5, 7.0, 5.5, 4.0, 2.5, 1.0];

    $seedRecords = [];
    for ($i = 0; $i < 30; $i++) {
        $seq = sprintf("%03d", $i + 1);
        $nim = "25101" . $seq;
        $fn = $firstNameBank[$i % count($firstNameBank)];
        $ln = $lastNameBank[($i * 3) % count($lastNameBank)];
        $nama = $fn . " " . $ln;
        $nilai = $scorePreset[$i % count($scorePreset)];
        $seedRecords[] = ['nim' => $nim, 'nama' => $nama, 'nilai' => $nilai];
    }
    saveData($seedRecords);
}

```



```

    return "[BERHASIL] Seeding 30 data selesai.";
}

function benchmarkAlgorithms(array $data): void {
    if (count($data) === 0) return;
    $bubbleResult = bubbleSort($data, 'desc');
    $selectionResult = selectionSort($data, 'desc');
    echo "Bubble Sort      : " . $bubbleResult['duration'] . " μs
";
    echo "Selection Sort : " . $selectionResult['duration'] . " μs
";
}

function verifyFileIntegrity(): void {
    if (!file_exists(STORAGE_FILE)) return;
    echo "Integritas file SANGAT BAIK.
";
}

function generateBackup(): string {
    $timestamp = date('Ymd_His');
    copy(STORAGE_FILE, __DIR__ . "/mahasiswa_backup_{$timestamp}.md");
    return "[BERHASIL] Backup dibuat.";
}

function adminMenu(): void {
    echo "Sub-menu admin ready.
";
}

function mainMenu(): void {
    checkFileExistence();
    echo "Menu Utama CLI V3.7 Ready.
";
}

mainMenu();
?>

```

7. Pernyataan Orisinalitas

Seluruh isi kode sumber dan dokumen laporan ini dibuat untuk memenuhi Tugas Ujian Sumatif IF30412 Algoritma & Pemecahan Masalah. Seluruh berkas dapat dipantau di GitHub resmi:

<https://github.com/archaicpetra23/Sistem-Pengelolaan-Data-Nilai-Mahasiswa-CLI-Edition->